Part A. PERSONAL INFORMATION

CVA date	09/11/2026

First name	Rodrigo		
Family name	Granjel Rodríguez		
Gender	Male	Birth date	09/07/1993
ID number	53817055E		
Email	rodrigo.granjel@bc3research.org	URL Web	https://granjel.github.io/
Open Researcher and Contributor ID (ORCID)		0000-0002-7785-4106	

A.1. Current position

Position	Postdoctoral researcher		
Start date	15/05/2023 (42 months*)		
Institution	BC3 – Basque Centre for Climate Change		
Department	Ecosystems and Global Change		
Country	Spain	Phone number	+34 667 04 86 44
Keywords	Community ecology; Species coexistence; Biotic interactions; Global change; Ecosystem stability and recovery		

*Including 5 months of childcare leave in separate periods between 10/2025 and 09/2026.

A.2. Previous positions

Period	Position / Institution / Country
06/2022 – 03/2023 (10 months)	Fulbright research fellow / North Carolina State University / USA
11/2017 – 03/2022 (53 months)	FPU doctoral researcher / Universidad de Sevilla / Spain
09/2017 – 10/2017 (2 months)	JAE-INTRO research assistant / IRNAS-CSIC / Spain

A.3. Education

Degree / Master's / PhD	University / Country	Year
PhD in Integrative Biology	Universidad de Sevilla / Spain	2023
MSc in Terrestrial Ecology	Universitat Autònoma de Barcelona / Spain	2017
BSc in Biology	Universidad de Vigo / Spain	2016

Part B. CV SUMMARY

I investigate how species interactions determine which species can persist together under environmental change. I combine detailed empirical data from my own field and laboratory experiments and international collaborations with mathematical models grounded in coexistence theory and ecological network approaches. My work lies at the intersection of experimental and quantitative ecology.

Scientific contributions and initiative: During my FPU-funded PhD at the University of Sevilla, I linked nitrogen enrichment and fungal pathogens to the mechanisms of multispecies coexistence. With Eric Allan in Bern, I showed that the number of interacting species changes the importance of direct and indirect effects on coexistence (*New Phytologist*, 2023). I co-designed the research, conducted fieldwork, analysed the data and wrote the manuscript. In *Oikos* (2026), I led the analyses and software development linking simple plant interaction patterns to structural niche and fitness differences in a multitrophic grassland community. My experimental work in chemical ecology showed that volatile signalling reduces herbivore damage in *Arabidopsis* populations with contrasting life histories (*Environmental and Experimental Botany*, 2026). I am **first and corresponding author of these three papers**. For the latter research, I conceived and led CoVOCom, securing an AEET early-career project grant and coordinating the experiments.

Sustained international collaboration: My research visits and fieldwork span six countries outside Spain. Repeated visits to Bern supported my grassland research and a continuing collaboration, including a first-author manuscript on land-use intensification and coexistence, currently under review. Following my stay with Tadeu Siqueira at UNESP, Brazil, I continue to collaborate with Brazilian researchers on agrochemical effects on planktonic food webs, with a manuscript in preparation. As a Fulbright fellow at North Carolina State University, USA, I conceived and conducted a project on microbial interactions, stability and functioning, hosted by William Petry. Our collaboration continues through a manuscript in preparation testing predictions of competitive outcomes. Earlier fieldwork at the Swedish University of

Agricultural Sciences broadened my experience of above- and below-ground responses to disturbance. These relationships allow me to test ecological ideas across systems and sustain joint research beyond individual visits.

Current research: Since 2023, I have worked with Daniel Montoya at BC3 within the ERC project RECODYN. Alongside fieldwork at CNRS SETE's terrestrial Metatron (Moulis, France), I am developing Bayesian state-space models to infer plant growth and interactions from biomass time series, accounting for observation and process uncertainty. With Jürg Spaak, whom I visited at RPTU, Germany, in 2026, I connect these estimates to coexistence theory and assembly graphs to investigate recovery pathways under warming. I also contribute experiments and field sampling to ChEMISt, linking chemical ecology with multitrophic interactions. I contributed analyses, interpretation and writing to a study distinguishing local and spatial contributions of plant diversity and functional traits to productivity stability across 235 grasslands (*Nature Communications*, 2026). Current collaborative work includes a manuscript on warming and recovery under review and another on intraspecific trait variation and stability in preparation. Together, these studies extend my coexistence research towards understanding how functional differences influence ecosystem recovery.

Open science, training and service: I share data and R code through public repositories, including the PaNDiv archive and software for interaction estimation and structural coexistence analyses. I have taught 192 hours at undergraduate and master's level, co-supervised an MSc thesis and mentored two DAAD RISE Worldwide internships, one including BSc co-supervision. I serve on the PCI Ecology recommenders panel, have evaluated Fulbright fellowships and AEET projects, and served on the scientific committee and coordinated a session at the 2025 SIBECOL–AEET meeting. Through the European Researchers' Night and school workshops, I share ecological research with wider audiences.

Next research step: Working with Carlos Pérez Carmona, I aim to combine my experimental and interaction-modelling experience with his expertise in functional ecology to test when functional variation predicts coexistence and recovery under environmental change.

Part C. RELEVANT MERITS

C.1. Publications

1. *Journal article:* **Granjel RR**, Gross N, Badenhausser I, Arroyo J, Godoy O. 2026. Simple plant–plant interaction patterns predict structural niche and fitness differences in a multitrophic grassland community. OIKOS, e12212 doi: 10.1002/oik.12212

   > Corresponding author and position 1/5 authors. I used statistical models to quantify key direct and higher-order interactions among plants and grasshoppers. We show that simple patterns of competition and facilitation between plants help explain coexistence within a complex food web.

2. *Journal article:* **Granjel RR**, Martín-Cacheda L, Röder G, Izquierdo-Ferreiro I, Martín-Díaz A, Picó FX. 2026. Plant–plant signalling reduces herbivore damage in two *Arabidopsis thaliana* populations with contrasting life histories. ENVIRONMENTAL AND EXPERIMENTAL BOTANY 251, 106472 doi: 10.1016/j.envexpbot.2026.106472

   > Corresponding author and position 1/6 authors. From my AEET-funded project. We show that volatile communication increases resistance to herbivory in two contrasting populations of *A. thaliana*, and suggest that herbivory may reshape links between chemical and life-history traits.

3. *Journal article:* Huang M, **Granjel RR**, Montoya D, Xiao Y, Liu X, Hautier Y. 2026. Plant diversity within communities, not among them, stabilizes grassland productivity across spatial scales. NATURE COMMUNICATIONS 17, 2145 doi: 10.1038/s41467-026-69028-5

   > Position 2/6 authors. I contributed to statistical analyses, interpretation and writing. We show that functional traits mediate the relationship between environmental conditions and ecosystem stability differently across spatial scales.

4. *Journal article:* Montoya D, Burón-Ugarte A, …, **Granjel RR**, …, Porras-Gómez J. 2025. Complexity and interpretability in global change ecology. PLOS CLIMATE 4, e0000587 doi: 10.1371/journal.pclm.0000587

   > Position 5/10 authors. First contribution of my postdoc at BC3. I helped conceptualise and write this piece on the complexity–interpretability trade-off in global change experiments.

5. *Journal article:* Jiménez-Mejías P, Manzano S, …, **Granjel RR**, …, Fitzpatrick M. 2024. Protecting stable biological nomenclatural systems enables universal communication: A collective international appeal. BIOSCIENCE 74, 467–472 doi: 10.1093/biosci/biae043
   > Position 1280/1563 authors. I endorsed this international appeal for stable biological nomenclature and universal scientific communication.
6. *Journal article:* **Granjel RR**, Allan E, Godoy O. 2023. Nitrogen enrichment and foliar fungal pathogens affect the mechanisms of multispecies plant coexistence. NEW PHYTOLOGIST 237, 2332–2346 doi: 10.1111/nph.18689
   > Corresponding author and position 1/3 authors. My first PhD paper, where I led all steps from data acquisition in Switzerland to manuscript preparation. We show that the effects of nutrients and pathogens on coexistence mechanisms need to be understood beyond pairwise interactions.
7. *Journal article:* Moreira X, **Granjel RR**, …, Rasmann S. 2021. Apparent inhibition of induced plant volatiles by a fungal pathogen prevents airborne communication between potato plants. PLANT, CELL & ENVIRONMENT 44, 1192–1201 doi: 10.1111/pce.13961
   > Position 2/10 authors. Derived from a BSc thesis I co-supervised at LMU Munich. We show that fungal infection determines how potato plants communicate via chemical volatiles.
8. *Journal article:* Juan-Ovejero R, **Granjel RR**, Ramil-Rego P, Briones MJI. 2020. The interplay between abiotic factors and below-ground biological interactions regulates carbon exports from peatlands. GEODERMA 368, 114313 doi: 10.1016/j.geoderma.2020.114313
   > Position 2/4 authors. Derived from my MSc thesis. We link abiotic conditions and interactions among plants, soil fauna and microbes to a clear peatland ecosystem function: carbon exports.
9. *Book chapter:* **Granjel RR**, Juan Ovejero R, Briones MJI. 2017. Diversidad edáfica en diferentes hábitats turbosos de la serra do Xistral. CONSERVACIÓN E XESTIÓN DE HUMIDAIS EN GALICIA. HORREUM-IBADER, 31–48. ISBN: 978-84-939661-4-0
   > Position 1/3 authors. Derived from my BSc thesis. It represents the first characterisation of soil biodiversity across different peatland habitats in Serra do Xistral, Galicia.

C.2. Conferences

1. *Invited seminar:* **Granjel RR**. Starting over: Or the hard life of plants in a recovery dynamics experiment. Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau (Landau, Germany; June 2026).
2. *Scientific committee member, session coordinator and moderator:* The interplay between species coexistence and abiotic factors: community ecology (general session GSC). III SIBECOL & XVII AEET Meeting (Pontevedra, Spain; June 2025).
3. *Poster presentation:* **Granjel RR**, Montoya D. How does warming affect the coexistence mechanisms of recovering communities? III SIBECOL & XVII AEET meeting (Pontevedra, Spain; June 2025).
4. *Invited seminar:* **Granjel RR**. `Theory < data` for community ecology. Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau (Landau, Germany; April 2025).
5. *Oral presentation:* **Granjel RR**, Buche L, Spaak JW, Godoy O. Environmental variability and the mechanisms behind species persistence. XV AEET National Conference (Plasencia, Spain; October 2021).
6. *Oral presentation:* **Granjel RR**, Allan E, Godoy O. The consequences of nitrogen enrichment and foliar pathogens for multispecies coexistence in grasslands. Festival of Ecology BES (Online due to the COVID-19 pandemic; December 2020).
7. *Poster presentation:* **Granjel RR**. Coexistence and VOC-mediated plant–plant communication in response to insect herbivory (CoVOCom). XVII ECOFLOR Meeting AEET (Bilbao, Spain; March 2020).
8. *Invited seminar:* **Granjel RR**. The struggle for coexistence: Studying biodiversity beyond pairwise interactions. Universität Bern (Bern, Switzerland; October 2019).
9. *Oral presentation:* **Granjel RR**, Gross N, Godoy O. How does the interaction network determine the herbivory effects on plant coexistence? 1st SIBECOL Meeting (Barcelona, Spain; February 2019).

10. *Poster presentation:* **Granjel RR**, Arroyo J, Godoy O. How do biotic interactions and environmental conditions shape species coexistence? Ecology Across Borders 2017 (Ghent, Belgium; December 2017).

C.3. Research projects

1. ChEMISt: Unveiling the factors shaping agroecosystem stability under global change: phytochemistry, functional traits, and multitrophic interactions. Ref. 24-C1-Agro-TNA-032. AgroServ Transnational Access, Horizon Europe. Co-PIs: Lucille T. S. Chrétien and Daniel Montoya, at BC3 (Spain); 2024–2026. Estimated value: €85,000. Awarded access to facilities and analytical services: CNRS-SETE (France) and CZU (Czech Republic). Ongoing.
   > As a beneficiary and postdoctoral researcher, I implement experiments and field sampling linked to RECODYN.
2. RECODYN: Ecosystem recovery dynamics and their response to climate change and habitat fragmentation. Ref. 101043548. ERC Consolidator Grant, European Research Council. PI: Daniel Montoya, at BC3 (Spain); 2023–2027. Total funding: €1,999,315. Ongoing.
   > As a postdoctoral researcher, I conduct experiments, field sampling, data analysis and Bayesian state-space modelling of interactions, coexistence and community recovery under warming.
3. SISMiC: The role of simple interactions in the stability and function of microbial communities. Predoctoral Research Fellowship, Fulbright Spain and Regional Government of Andalusia. PI: Rodrigo R. Granjel, at North Carolina State University (USA; William Petry's group); 2022–2023. Total funding: US$23,660. Completed.
   > I conceived this project to understand how simple assembly rules lead to complex, functional ecosystems, using experiments with microbial communities in the lab.
4. CoVOCom: Coexistence and chemical communication between plants in response to insect herbivory. Early-career Ecology Project Grants (“Tomando la iniciativa”), Spanish Association of Terrestrial Ecology (AEET). PI: Rodrigo R. Granjel, at Universidad de Sevilla and MBG-CSIC (Spain); 2020. Total funding: €2,000. Completed.
   > My first project as an independent PI. It combined my two main interests, interactions and coexistence, to understand how plants persist under herbivory via chemical communication.
5. MeDiNaS: Mechanisms of biodiversity maintenance in variable environments. Ref. RTI2018-098888-A-I00. Plan Nacional de I+D+i / Retos Investigación, Spanish Ministry of Science, Innovation and Universities. PI: Oscar Godoy, at Universidad de Cádiz (Spain); 2019–2021. Total funding: €105,270. Completed.
   > Led by my PhD supervisor. I contributed to fieldwork and discussions as a work-team member.

C.4. Knowledge and technology transfer

I developed software for estimating species interactions using lasso regularisation and calculating structural niche and fitness differences, released under the MIT licence (2026; doi: 10.5281/zenodo.23054165).

In 2019, I provided a biological expert assessment for judicial proceedings before the court in Padrón (Galicia, Spain), addressing environmental and potential public health concerns associated with urban wildlife in a residential building.